Yixiong Sun

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PROFESSIONAL SUMMARY

Systems neuroscience PhD candidate specializing in Alzheimer's disease, integrating experimental neuroscience with neural engineering, machine learning, and computational modeling. Leads end-to-end research on neural circuits and memory, spanning longitudinal recordings, closed-loop stimulation, large-scale data analysis, and mechanistic simulations. Applies production software engineering experience to develop and validate integrated experimental and computational systems. Coordinates interdisciplinary projects, mentors researchers, and communicates complex scientific and technical findings to diverse audiences.

EDUCATION

University of Toronto — Toronto, ON
PhD, Cell and Systems Biology (Systems Neuroscience) — Sep 2021 – Expected 2027
Thesis: Linking abnormal hippocampal neural activity and memory deficits to Aβ accumulation in Alzheimer's disease
GPA: 4.0/4.0

McGill University — Montreal, QC
Bachelor of Science, Joint Honours Computer Science and Biology — Sep 2017 – Apr 2021
GPA: 3.96/4.0

EXPERIENCE

Takehara Lab, University of Toronto — Toronto, ON
PhD Researcher, Systems Neuroscience — Sep 2021 – Present

Data Sciences Institute Doctoral Student Fellowship (2024–2027)
Competitively selected as 1 of 14 recipients · $75,000 total funding

- Designed and led an investigation of hippocampal network dysfunction and memory impairment in Alzheimer's disease, involving 100+ mice and identifying reduced ripple activity and impaired learning-related reorganization of neuronal populations.
- Conducted longitudinal neural recording studies in freely moving mice, spanning custom microdrive fabrication, stereotaxic implantation, single-unit/LFP acquisition, multi-day behavioral and sleep experiments, and histological analysis.
- Provided the first evidence of a candidate neuroprotective peptide's efficacy in a chronic Alzheimer's mouse model, rescuing spatial-memory deficits and impaired learning-related neuronal reorganization while increasing hippocampal ripple activity.
- Engineered and validated closed-loop stimulation systems in Alzheimer's-model mice, integrating hardware-triggered motor-cortex stimulation following hippocampal ripples (20-ms delay) to induce cortical slow oscillations, and software-mediated theta phase-locked stimulation of CA3 inputs to CA1 (±10-ms accuracy).
- Built automated Python and MATLAB pipelines to process 4 TB of electrophysiology, behavioral, and imaging data, supporting neural-event detection, population-level analyses, statistical testing, and visualization.
- Extended a hippocampal spiking network model to reproduce experimental neural activity and test competing mechanisms of impaired learning-related network adaptation in Alzheimer's disease.
- Mentored nine undergraduate researchers, trained six graduate students in electrophysiology and data analysis, and independently led biology and animal physiology laboratory sessions as a teaching assistant.

Waldispühl Lab, McGill University — Montreal, QC
Undergraduate Researcher — Sep 2020 – Apr 2021

- Extended and validated BayesPairing2, an RNA secondary-structure and motif discovery pipeline, optimizing Bayesian-network construction and RNA model loading.

Beagle Inc — Kitchener, ON
Software Developer — May 2021 – Aug 2021

- Improved document-parsing and machine learning pipelines for legal text recognition, integrating supervised classification and iterative data labeling.
- Modernized and containerized a legacy Django platform, implementing CI/CD and deploying to AWS infrastructure.

iVenuto.com Corporation Kitchener, ON
Software Engineer, Consultant May 2018 – Aug 2021

- Owned the architecture and full-stack development of a commercial visitor-management platform serving 100+ customers, implementing features including video calling, offline synchronization, and mobile integration.
- Engineered CI/CD pipelines and managed AWS-hosted staging and production infrastructure, including Linux/Nginx servers, databases, and background workers; delivered 10+ software releases.
- Led feature planning and code reviews, mentored junior developers, and resolved complex production issues escalated by customer support.

SELECTED TECHNICAL PROJECTS

EvidNeuro – Evidence-Grounded Neuroscience Retrieval Engine Aug 2026 – Present
TypeScript, LLMs, MinerU, PostgreSQL/pgvector

- Architected a scientific literature retrieval system that connects evidence across neuroscience papers, extracting structured claims, measurements, and biological relationships while preserving links to original sources.
- Implemented multi-layer retrieval combining BM25/vector search, LLM reranking, and knowledge-graph queries to support cross-paper evidence discovery and retrieval of related scientific findings.
- Processed 60 neuroscience papers into 15,000+ evidence objects and 9,000+ structured claims; achieved 99.05% precision and 99.88% recall in selecting relevant scientific prose during PDF postprocessing.

Deep Learning for Neural Event Classification Jan 2026 – Present
Python, PyTorch, signal processing, CNNs, subject-aware evaluation

- Developed an end-to-end deep learning pipeline to distinguish physiological hippocampal ripples from pathological epileptiform discharges (IEDs) and noise, using 7,701 labeled LFP events from 30 mice across control, Alzheimer's-model, and treatment groups.
- Designed a multi-representation CNN integrating multichannel waveforms, time-frequency spectrograms, cross-channel attention, and coherence- and entropy-based features to characterize neural event morphology and oscillatory structure.
- Achieved 0.849 macro F1 and 86.1% accuracy in subject-grouped cross-validation (5 folds × 3 seeds), outperforming an RBF-SVM baseline (0.753 macro F1, 77.1% accuracy) in all 15 paired evaluations.

Mechanistic Modeling of Hippocampal Network Dysfunction Sep 2024 – Dec 2025
Python, Spiking Neural Networks, Computational Neuroscience, Brian2

- Extended a CA3–CA1 spiking network model with experience-dependent synaptic strengthening, reproducing enhanced ripple-associated firing in 68% of CA1 pyramidal neurons (vs. 70% experimentally) and learning-related changes in neuronal coactivity.
- Tested competing excitatory and inhibitory circuit dysfunctions against experimental findings, identifying elevated synaptic excitation as a better explanation for impaired neuronal recruitment and network reorganization in Alzheimer's disease.

TECHNICAL SKILLS

Neuroscience and Neural Engineering: *In vivo* single-unit/LFP recording, microdrive fabrication and implantation, closed-loop stimulation, neural recording systems (Neuralynx, Open Ephys), behavioral experiments, hardware/software integration
Machine Learning and AI: PyTorch, scikit-learn, CNNs, cross-validation, subject-aware evaluation, LLM pipelines, hybrid retrieval, knowledge graphs
Scientific Computing: Python, MATLAB, NumPy, SciPy, pandas, signal processing, time-frequency analysis, statistical modeling, network analysis, Brian2
Software Engineering: TypeScript/JavaScript, Node.js, C/C++, Django, PostgreSQL, MongoDB, Docker, AWS, CI/CD, Linux, Git

SELECTED PUBLICATIONS AND PRESENTATIONS

Sun Y, Chekhov S, Zhang K, et al. Sharp wave-ripple disruption limits experience-dependent neuronal ensemble reorganization in amyloid-associated memory impairment. *Manuscript under review at Current Biology*, 2026.

Sun Y, Takehara-Nishiuchi K. The medial prefrontal cortex leaves the hippocampus when it prepares for the future. Science Progress, 2024.

Sun Y, Taxidis J, Takehara-Nishiuchi K. In silico modeling of amyloidosis-induced CA1 spike abnormalities: Unraveling the role of excess synaptic glutamate. Society for Neuroscience, Poster, 2025.